TEXORA · LECTURE SHEET

Warp Knitting Calculations

Warp knitting counts differently: the machine is fed from beams, yarn consumption is booked by the rack of 480 courses, and each guide bar has its own appetite. This sheet works the tricot calculation end to end — production rate, run-in, yarn per square metre, GSM — and the beam preparation arithmetic behind it.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 11 · SPECIALIZATION — FABRIC MANUFACTURING

1. The machine and its units

PARAMETER	VALUE (WORKED EXAMPLE)
Machine	Two-bar tricot, 28 gauge, working width 152 in (3.86 m)
Speed	1,500 courses/min
Structure	CPC 24 per cm in the finished fabric
Yarn	Polyester filament 78 dtex (7.8 tex), both bars
Run-in, bar 1 (back)	1,450 mm per rack
Run-in, bar 2 (front)	1,150 mm per rack

Run-in is the length of yarn one end of a guide bar delivers while the machine knits one **rack** — 480 courses, the warp knitter's standard counting unit. Run-in, not stitch length, is the warp knitter's consumption variable: it is set by the let-off and measured on the machine, bar by bar.

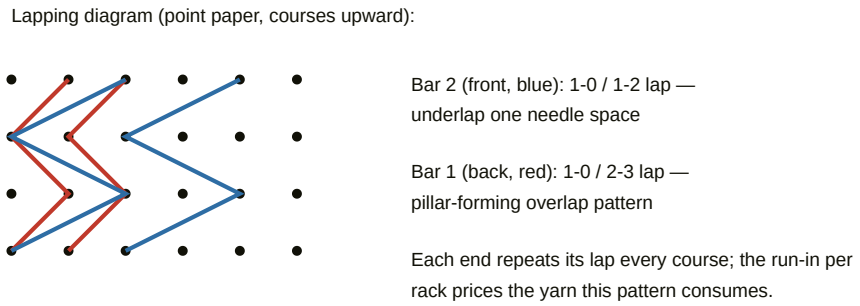


Figure 1 — The two laps of the worked example on point paper: the pattern each bar repeats, course after course, rack after rack.

2. Production rate

Fabric length per hour = courses per minute × 60 ÷ courses per metre

WORKED EXAMPLE — TRICOT OUTPUT

Length = 1,500 × 60 ÷ (24 × 100) = **37.5 m/h** at 3.86 m width — 144.8 m² of fabric per hour. Warp knitting is fast in courses and modest in metres: the CPC of a warp knit is high, and the two statements of output (courses and metres) must never be swapped in a report.

3. Yarn consumption and GSM — by the rack

WORKED EXAMPLE — CONSUMPTION OF THE TWO BARS

Ends per bar = gauge × width in inches = 28 × 152 = **4,256 ends**.

One rack = 480 courses produces 480 ÷ 24 = 20 cm = 0.20 m of fabric; area per rack = 0.20 × 3.86 = **0.772 m²**.

Yarn per rack: bar 1 = 4,256 × 1.45 m = **6,171 m**; bar 2 = 4,256 × 1.15 m = **4,894 m**; total **11,066 m per rack**.

Yarn per m² = 11,066 ÷ 0.772 = **14,334 m/m²**.

GSM = 14,334 × 7.8 tex ÷ 1,000 = **112 g/m²** — a typical tricot weight, obtained without weighing anything.

Run-in ratio bar 1 : bar 2 = 1,450 : 1,150 = **1.26** — the back bar works harder, as pillar-forming bars do.

The same rack arithmetic prices a beam: yarn per beam set = run-in × ends × racks in the beam length. A beam carrying 20,000 m of fabric length holds 20,000 ÷ 0.20 = 100,000 racks, so bar 1's beam must carry 6,171 m × 100,000 ÷ 1,000... in weight: 100,000 racks × 6,171 m × 7.8 tex ÷ 10^6 = **4,813 kg** — which is why warp-knitting beams are sectional and numerous rather than single and huge.

4. Beam preparation (sectional warping for warp knitting)

Warp-knitting warps are wound as sections on a sectional warper, one section per beam of the set. With 4,256 ends per bar prepared in 8 sections, each section carries 4,256 ÷ 8 = **532 ends** — the creel and section-width arithmetic of Sheet 2 in a finer gauge. The non-negotiable requirement is equality: every section the same length at the same tension, because all sections of a bar are drawn off together and a 0.1% length difference arrives at the machine as a pattern of tight and slack ends that no run-in setting can remove.

Glossary

Rack — 480 courses, the warp-knitting counting unit. **Run-in** — yarn length delivered per end per rack, per guide bar. **Run-in ratio** — relative consumption of two bars. **Lapping** — the guide bar's movement pattern (overlap + underlap) that forms the structure. **Sectional beam** — one of a set of beams whose ends combine into one guide bar's warp.

Key points

• Length rate = courses/min ÷ courses per metre (worked: 37.5 m/h at CPC 24).
• Consumption is booked per rack: ends × run-in, bar by bar (6,171 + 4,894 = 11,066 m/rack).
• GSM from run-in: 14,334 m/m² × 7.8 tex → 112 g/m² — no weighing required.
• Warp-knitting warps are sectional (532 ends/section here) and must be equal in length and tension.

Practice

1. Find the length per hour of a machine at 1,800 courses/min producing CPC 20 per cm.
2. Bar run-ins are changed to 1,500 mm (bar 1) and 1,050 mm (bar 2) at the same ends and tex. Find the new GSM (area per rack unchanged at 0.772 m²).
3. Why is run-in, rather than stitch length, the controlled variable on a warp knitting machine?

Answer key

1. 1,800 × 60 ÷ 2,000 = **54 m/h**.
2. Yarn per rack = 4,256 × (1.50 + 1.05) = 10,853 m; per m² = 10,853 ÷ 0.772 = 14,058 m; GSM = 14,058 × 7.8 ÷ 1,000 = **110 g/m²**. (Total run-in fell, so the fabric lightened even though bar 1 alone rose — only the sum prices the cloth.)
3. Because the yarn is delivered from beams by a let-off, not drawn loop-by-loop from a package: what the machine controls is the length fed per rack. Stitch geometry follows from that feed together with the take-up; run-in is both the control knob and the consumption record.